

Docker: Lesson 01

The Problem

- **"It works on my machine!"** Developers often build apps that work locally but break when shared or deployed.
- This happens because different computers have different software, configurations, and hardware dependencies.

The Solution: Docker

Docker solves this by packaging the application and all its necessities into a lightweight, isolated environment.

- **Images:** Lightweight blueprints that contain the source code and all required dependencies.
- **Containers:** Stripped-down, virtualized environments created from images that run exactly what the app needs and nothing else.

Why It Matters

Because the container holds everything the app needs to function, it will run identically on *any* machine that has Docker installed. It completely eliminates environmental inconsistencies, making deployment faster and safer.

The Cooking Analogy: Imagine having a magic box that packs up your exact stove, pans, and ingredients so your family recipe tastes perfect no matter whose house you cook it in. That is what Docker does for software!